AI-Driven Predictive Hydrogen Leakage Prevention System (HyLeakAI)

Overview

As the world transitions toward clean energy, hydrogen has emerged as one of the most promising energy carriers for decarbonizing industries and balancing renewable power generation. One of the biggest challenges in adopting a hydrogen-based economy is the safe and economical storage of hydrogen at a large scale. Underground Hydrogen Storage (UHS) in depleted oil and gas reservoirs is increasingly being recognized as a viable solution because these reservoirs already possess the geological structures necessary for storing gases under high pressure.

However, hydrogen molecules are extremely small and highly diffusive, making them more susceptible to leakage through microscopic fractures, abandoned wells, or compromised caprocks. Existing monitoring systems are largely reactive, relying on pressure measurements, periodic inspections, and gas detection sensors that identify leakage only after it has already begun. Such delays can lead to significant hydrogen loss, safety hazards, environmental concerns, and substantial economic costs.

Our proposed solution, **HyLeakAI**, is an AI-driven predictive platform designed to detect the risk of hydrogen leakage before it occurs. Rather than simply monitoring reservoir conditions, the system continuously analyzes geological, operational, and historical data to forecast potential leakage zones, estimate safe operating pressures, and provide early warnings to operators. The objective is to transform underground hydrogen storage from a reactive process into a predictive and intelligent management system.

Proposed Solution

HyLeakAI integrates machine learning, subsurface intelligence, and reservoir engineering into a unified decision-support platform capable of evaluating the integrity of underground hydrogen storage reservoirs in real time.

The platform utilizes publicly available datasets such as the **Equinor Volve Field Dataset**, **Dutch F3 Seismic Dataset**, well-log datasets, and open seismic repositories. These datasets contain valuable geological information including 3D seismic surveys, rock properties, well logs, reservoir pressure histories, and fault maps that serve as the foundation for training predictive AI models.

The proposed system consists of four major components:

1. Geological Intelligence Module

This module analyzes seismic images, well logs, porosity, permeability, caprock thickness, and historical production data to understand the structural characteristics of the reservoir. Machine learning models estimate the stability of the caprock, identify fault zones, and assess the overall geological suitability of the storage site.

2. AI-Based Leakage Prediction Engine

This is the core innovation of the project.

Using advanced machine learning techniques such as Long Short-Term Memory (LSTM) networks, Graph Neural Networks (GNNs), Physics-Informed Neural Networks (PINNs), and ensemble learning models, the system predicts the probability of hydrogen leakage under different operating conditions.

The model continuously evaluates parameters including:

- Reservoir pressure
- Injection rate
- Temperature
- Rock permeability
- Rock porosity
- Fault proximity
- Historical seismic activity

Based on these inputs, the AI predicts:

- Leakage probability
- Safe injection pressure limits

- Potential leakage pathways
- Reservoir health score
- Early warning alerts

Instead of detecting failures after they occur, the model enables preventive action by forecasting unsafe operating conditions before hydrogen injection becomes risky.

3. Digital Twin & Visualization Dashboard

To improve decision-making, the system creates a digital twin of the underground reservoir. Engineers can visualize reservoir conditions through an interactive dashboard displaying:

- Three-dimensional reservoir structure
- Hydrogen migration pathways
- Leakage probability heatmaps
- Pressure distribution
- Fault activation zones
- Safe operating boundaries

This visualization enables operators to understand complex underground processes intuitively and take timely corrective actions.

4. Economic & Operational Optimization

The final module evaluates the economic feasibility of underground hydrogen storage using predictive analytics.

The platform estimates:

- Operational costs
- Hydrogen losses due to leakage
- Compressor energy consumption
- Storage efficiency
- Return on investment (ROI)

Additionally, time-series forecasting models can recommend optimal hydrogen injection and withdrawal schedules based on renewable electricity availability and market electricity prices, thereby maximizing profitability while ensuring operational safety.

Expected Deliverables

The project aims to develop an integrated prototype consisting of:

- An AI model capable of predicting hydrogen leakage before it occurs.
- A Reservoir Health Score that continuously evaluates storage integrity.
- A digital twin dashboard providing real-time visualization of reservoir conditions.
- Leakage probability heatmaps and risk assessment reports.
- Safe injection pressure recommendations.
- An economic analysis demonstrating the financial benefits of predictive leakage prevention compared to traditional monitoring systems.

Innovation and Impact

The primary innovation of HyLeakAI lies in shifting underground hydrogen storage from **reactive monitoring** to **predictive intelligence**. Existing systems detect leakage only after hydrogen has escaped, whereas our platform forecasts leakage risk before unsafe conditions develop.

By combining geological data, machine learning, digital twin technology, and economic optimization into a single AI-powered framework, the proposed solution offers a comprehensive decision-support system for safe and efficient underground hydrogen storage.

Successful implementation of this platform can significantly reduce hydrogen losses, improve storage safety, lower operational costs, and accelerate the deployment of hydrogen as a sustainable energy carrier. Beyond hydrogen storage, the framework can be extended to carbon sequestration, compressed air energy storage, and geothermal reservoir management, making it a versatile solution for future subsurface energy applications.